Part I
Encryption and cryptography

1 Principles

Vocabulary

- Cyphering / Encrypting
- Cyphertext / Plaintext
- Cryptography
- Cryptanalysis

How to design robust algorithms?

Formal methods

- Complexity theory
- Information theory

Generic objectives

1. Easy encryption/decryption
2. Hard code-breaking
3. Strictness about organisational methods
4. Sensible assessing about the adversary

Useful tips

- Add some redondancy
- But not too much

- Distinguish the old messages from the new ones

Algorithms and cyphering keys

- Algorithms use keys
- Key's length linked to security level
- Robustness based on algorithm's strength and secure storage of keys
- Systems should be *computational secure*

Cryptanalysis

- Brute-force attack
- Related-key attack
- Chosen-plaintext/chosen cyphertext attack
- Collisions

- Testing keys with a few different bits (AES)

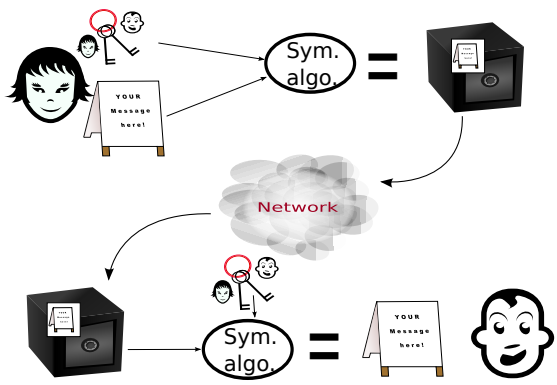
Basic coding methods

- Substitution
- Transposition
- One-Time Pad

2 Standard cryptography

2.1 Symmetric cryptography

Symmetric cryptography

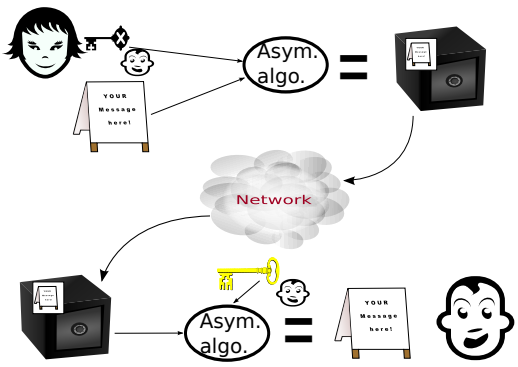


Standard algorithms

- Data Encryption Standard (and successors)
- RC2, RC4, RC5
- International Data Encryption Algorithm
- Digital Signature Algorithm
- Advanced Encryption Standard

2.2 Asymmetric cryptography

Asymmetric cryptography



Operating mode

- Encryption key: **public** / decryption key : **private**
- Encryption / Decryption: **RSA**, **Diffie-Hellman**, **El-Gamal**, **Elliptical curves**...
- (*Hashing*) documents → **MD5**, **SHA-1**, **SHA-2**, **SHA-3**

Theoretical issues

- Discrete logarithm on a finite field

→ find an integer x, $0 \leq x \leq p-2$, given an integer y such that $g^x \mod p = y$.

- Discrete logarithm over an elliptical curve
  - Curve $y^2 = x^3 + ux + v$ (u and v such that $4u^3 + 27v^2 \neq 0$) defined over any finite field
  - Operation: scalar product of a curve point with an integer i
  - Problem: assume the result of the operation and the curve point are known, find i

Diffie-Hellman key exchange

1. Alice and Bob: generator g / prime number n
2. Alice: secret integer $a \rightarrow g^a \mod n \rightarrow$ Bob
3. Bob: secret integer $b \rightarrow g^b \mod n \rightarrow$ Alice
4. Alice: $(g^b \mod n)^a \mod n$
5. Bob: $(g^a \mod n)^b \mod n$

RSA

Public key (e, n) p, q: private big integers and $n = pq$. Choose $e \rightarrow$ no divisor common with $(p-1)(q-1)$.

Private key d Find d such as $ed \mod (p-1)(q-1) = 1$

Encryption $C = M^e \mod n$.

Decryption $M = C^d \mod n$.

Potential weaknesses

- Mathematical methods to reverse coding of private keys
- Quantum computers could theoretically perform an exponential number of operations in one step

2.3 Key management

Time to live Key management

TTL depends on frequency of use

- Local file
- Network application

Management

- Multiple services
- System often based on physical devices

Public Key Infrastructure

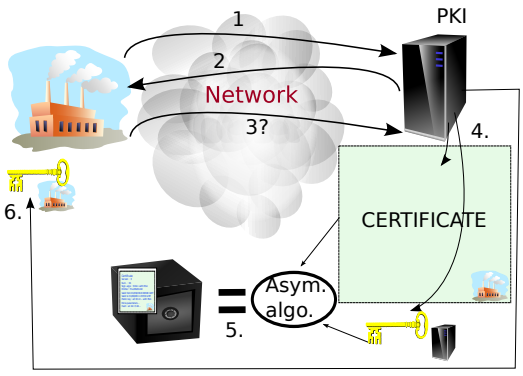
- Manage and distribute public keys
- Multiple services:
  - single pairs of keys (private/public)
  - digital certificates
  - certify public keys

Digital certificates

```
1.  Certificate version
2.  Serial number
3.  Algorithm used to hash the certificate
4.  Organization having issued the certificate
5.  Validity time
6.  Certificate's owner
7.  Owner's public key with algorithm
8.  Optional extra information
9.  Certificate hashing
```

PKI

Getting the certificate



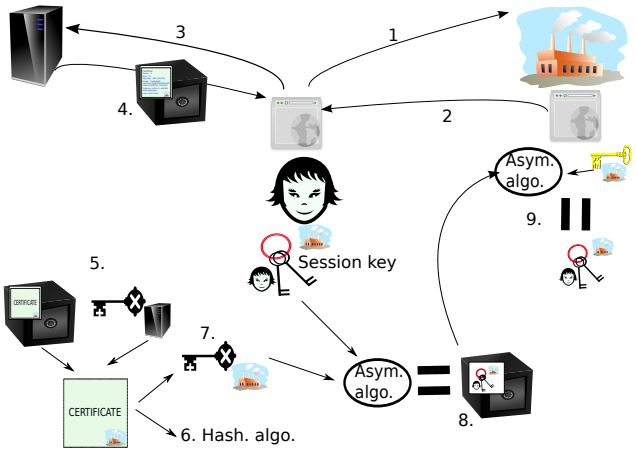
PKI

Using the certificate

Certificate revocation

- Managed by the PKI

→ publish a *Certificate Revocation List*



PKI limits

Implicit limits

- Complexity, management costs
- High security level
- Certificate overall management

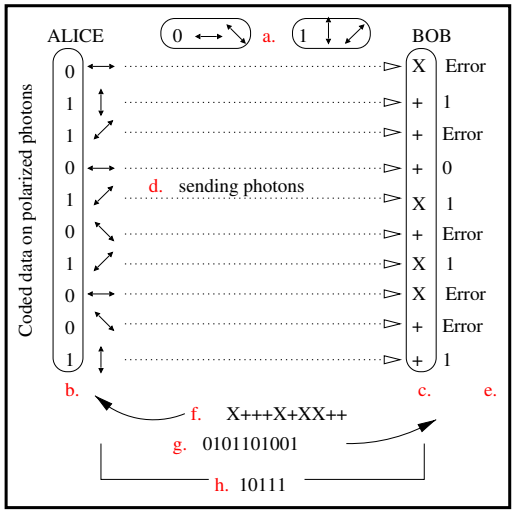
Trust-proof?

- Only one PKI → too much power

→ Some firms manage internal PKI

3 Quantum cryptography

Principle: Protocol BB84 (Brassard-Bennett) for quantum bits transmission



State of the art

Advantages

- Confidentiality
- Used as One-Time Pad
- **Computational secure** mechanism

Limits

- Currently limited
- Specific equipment
- Error correction

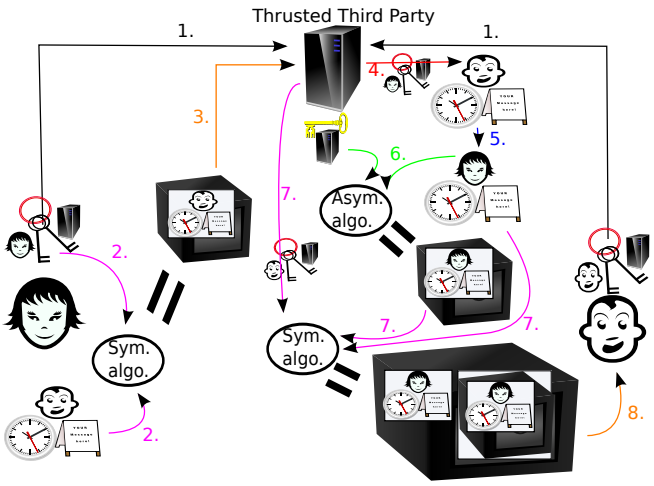
4 Cryptography-based services

Digital signing

3 conditions

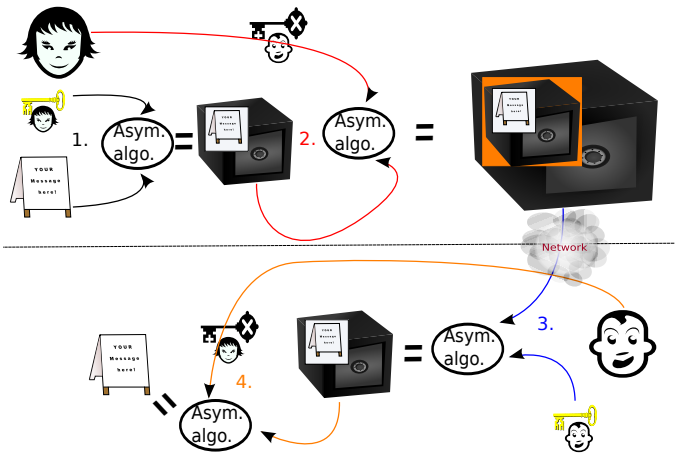
- Authentication
- Non-repudiation
- Non-forging

Signing with symmetrical keys



Signing with asymmetrical keys [

Condition: commutative private/public keys



Confidentiality vs. authentication

Confidentiality

- Encryption/decryption
- Keep private keys as long as possible:

Authentication

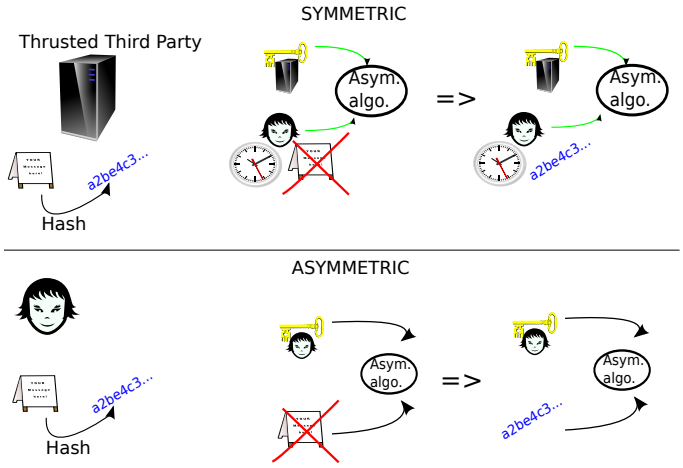
- Digital signature
- Keys to delete as soon as possible

Message digest

Principle

- Original message *not* encrypted
- **Hashing function**
  1. $\forall P$, $MD(P)$ is easy to compute.
  2. $\forall MD(P)$, no way to guess P.
  3. There is no P' such that $MD(P) = MD(P')$
  4. If P is modified in any way, $MD(P)$ changes a lot.

Message digest with encryption methods



Integrity and non-repudiation

Integrity

- Indecipherabledata can be deleted
- Computing time
- Collisions

Non-repudiation

Feasible with both asymmetric encryption and TTP